12. Ans: C

Since pressure is constant, thermodynamic temperature $T \propto$ volume V

i.e. $T = \left(\frac{p}{nR}\right)V$

$$\Rightarrow T/K = \theta/^{\circ}C + 273.15 = \left(\frac{p}{nR}\right)V$$

$$\theta/^{\circ}C = \left(\frac{p}{nR}\right)V - 273.15$$

This equation explains the negative Y-intercept.

So y-axis is temperature in °C and x-axis is volume in m^3.

13. Ans: A